TECHNICAL ARCHITECTURE & TRAINING REPORT

IMAGE-ONLY FULL UPPER-BODY AI MODEL & RECOGNITION PIPELINE

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1. Executive Summary & System Objectives

This technical document provides a comprehensive blueprint and code architecture explanation for the **IMAGE-ONLY Full Upper-Body AI Model Pipeline**. The system is engineered to perform high-accuracy 3D landmark detection, scale-invariant body-relative normalization, 3D interior joint angle calculations, and upper-body posture/gesture recognition strictly using **static images**.

STRICT 100% IMAGE-ONLY POLICY SATISFACTION:
• ZERO video files or temporal video processing.
• NO video frame extraction or temporal sequence models.
• Complete dataset creation, quality control, annotation, train/val/test splitting, augmentation, model training, evaluation, and single-image testing are executed strictly on static image matrices.

Target Upper-Body Anatomy Recognized:

- **Head & Facial References:** Nose, Left/Right Eyes, Left/Right Ears
- **Shoulder Belt:** Left Shoulder, Right Shoulder, Shoulder Alignment & Slope
- **Arms & Elbows:** Left/Right Upper Arms, Left/Right Elbow Joints, Left/Right Forearms
- **Wrists & Hands:** Left/Right Wrists, Left/Right Hand Regions (Pinky, Index, Thumb)
- **Torso & Spine:** Chest Region, Spine Alignment, Torso Tilt Angle relative to vertical
- **Hip References:** Left Hip, Right Hip, Pelvic Anchor Points

2. Technical Stack & Python Libraries Used

The system leverages industry-standard open-source machine learning and computer vision frameworks. Below is a detailed breakdown of each Python library and its specific role in the pipeline:

Library	Version	Specific Role & Pipeline Functionality
OpenCV (cv2)	4.13.0	Static image loading, BGR/RGB color space conversions, image blurring, noise filtering, geometric drawing (skeleton connections, joint keypoint circles), and visual output export.
MediaPipe (mediapipe)	0.10.14	Deep-learning pose landmarker solution (`mp.solutions.pose.Pose`). Extracts 33 keypoints in 3D (X, Y, Z) normalized coordinate space with presence and visibility scores.
NumPy (numpy)	2.2.6	High-performance 3D vector geometry, dot products, vector magnitudes, trigonometric arccos calculations, array slicing, and feature matrix preparation.

Scikit-Learn (sklearn)	1.9.0	Multi-candidate classifier training (`RandomForestClassifier`, `HistGradientBoostingClassifier`, `MLPClassifier`), `StandardScaler`, `LabelEncoder`, cross-validation, and metrics.
Pillow (PIL)	11.3.0	Procedural 2D/3D human anatomical drawing, limb capsule rendering, polygon drawing, color adjustments, and synthetic image synthesis.
Joblib (joblib)	1.4.2	Serialization and binary export of trained model weights, feature scalers, and label encoders to disk (`models/final/*.pkl`).
PyYAML (pyyaml)	6.0.2	Central configuration parsing and management (`config.yaml`).

3. How the Model Recognizes & Analyzes Upper Body

The system executes a 7-stage sequential processing workflow for every static input image. Below is the exact step-by-step mathematical and computational pipeline:

Step 1: Image Acquisition & Preprocessing

The static image is read via OpenCV as a BGR numpy matrix and converted to RGB color space required by MediaPipe.

Step 2: MediaPipe 3D Landmark Extraction

MediaPipe Pose processes the RGB image to detect 33 anatomical keypoints. Each keypoint provides (x, y, z) coordinates and a visibility score.

Step 3: Body-Relative Normalization

To eliminate camera distance instability (where joints jump when a person moves close/far from the camera), coordinates are transformed relative to body center and body scale: S = (TorsoLength + ShoulderWidth) / 2.0. Normalized Point = (Point - ShoulderCenter) / S.

Step 4: 3D Interior Joint Angle Calculation

Computes interior angles at vertices using vector dot products: cos(theta) = (u . v) / (||u|| ||v||), theta = arccos(cos(theta)). Calculates Left/Right Elbow Angles, Left/Right Shoulder Elevation, Spine Tilt Angle, and Cross-Body Distances.

Step 5: Feature Embedding Vector Construction

Combines scale-normalized 3D coordinates and calculated joint angle features into a 1D numerical feature vector.

Step 6: Machine Learning Classification

Passes the standardized feature vector through the trained Multi-Candidate Classifier (Random Forest / MLP) to predict the pose category and confidence percentage.

Step 7: Skeleton Overlay & Visual Output Export

Draws green skeleton connection lines, red keypoint circles, and text metrics overlay onto the image, saving the output to 'output/test_annotated.jpg'.

Key Mathematical Formulas Used:

```
1. Shoulder Center Anchor:
    ShoulderCenter = (LeftShoulder + RightShoulder) / 2

2. Body Scale Factor:
    Scale Factor S = ( ||ShoulderCenter - HipCenter|| + ||LeftShoulder - RightShoulder|| ) / 2.0

3. Scale & Distance Invariant Coordinate Transformation:
    P_normalized = ( P_raw - ShoulderCenter ) / S

4. 3D Interior Joint Angle (Vertex B):
    u = A - B,  v = C - B
    Angle = arccos( (u . v) / ( ||u|| * ||v|| ) ) * (180 / pi)
```

4. Detailed File-by-File Code Breakdown

The project is structured in a modular architecture cleanly separating configuration, data generation, quality control, feature extraction, model training, evaluation, and inference:

• config.yaml *(Configuration File)*

Stores all global hyperparameters, directory paths (dataset/generated, dataset/train, models/final), quality control thresholds (min_pose_confidence: 0.15, phash_similarity_threshold: 5), landmark target lists, 12 pose classes, and split ratios (70% Train, 15% Val, 15% Test).

• src/dataset_generator.py *(Synthetic Image Generator)*

Contains SyntheticUpperBodyGenerator class. Uses procedural PIL drawing to render synthetic upper-body human figures across pose angles, camera framing (close 1.2m, medium 2.5m, far 4.0m), body yaw rotation (front, 3/4, side), lighting, skin tones, clothing, and background environments.

• src/real_human_dataset_fetcher.py *(Real Human Photo Fetcher)*

Contains RealHumanDatasetFetcher class. Fetches high-resolution photographs of real humans across 10 pose categories from curated public sources, extracts MediaPipe landmarks, calculates joint angles, and auto-annotates ground truth pose classes.

• src/quality_filter.py *(Safety & Quality Control Filter)*

Contains ImageQualityFilter class. Performs anatomical geometry checks (verifies shoulder/torso length > 0, no detached limbs), checks keypoint presence, and deduplicates images using 64-bit Perceptual Hashing (pHash). Rejections are saved to dataset/rejected/ with JSON logs.

• src/pose_detector.py *(MediaPipe Pose Wrapper)*

Contains PoseDetector class. Wraps MediaPipe Pose landmarker (mp.solutions.pose.Pose), extracts 33 keypoints with (x, y, z, visibility), calculates upper body confidence, and draws visual skeleton overlays.

• src/landmark_normalizer.py *(Landmark Normalizer Engine)*

Contains LandmarkNormalizer class. Computes ShoulderCenter and body scale S, transforms raw keypoints to camera-distance invariant coordinates, and extracts flattened 1D numerical feature arrays.

• src/angle_calculator.py *(3D Joint Angle Calculator Engine)*

Contains AngleCalculator class. Computes 3D interior vector angles for left/right elbow flexions, shoulder elevations, spine tilt relative to vertical, wrist cross-body radial distances, and bilateral symmetry deltas.

• src/annotation_generator.py *(Annotation & Pseudo-Labeler)*

Contains AnnotationGenerator class. Processes static images, extracts keypoints, normalizes coordinates, calculates angles, incorporates ground-truth fallbacks for synthetic samples, and saves structured JSON annotation files.

• src/augmentation.py *(Train-Set Augmenter Engine)*

Contains DatasetAugmenter class. Applies color jitter, noise, blur, and horizontal flipping ONLY to the Training set split. Correctly mirrors coordinates and swaps Left <-> Right landmark IDs and pose class labels.

• src/trainer.py *(Multi-Candidate Model Trainer)*

Contains UpperBodyModelTrainer class. Fits RandomForestClassifier, HistGradientBoostingClassifier, and MLPClassifier on standardized feature vectors. Compares validation performance and exports best model artifacts to models/final/.

• src/evaluator.py *(Evaluator & Optimization Loop)*

Contains UpperBodyEvaluator class. Evaluates models on unseen test data, calculates accuracy, precision, recall, F1, per-class accuracy, and joint angle MAE. Automatically detects weak pose classes and triggers targeted synthetic image generation.

• src/inference.py *(Inference Engine Module)*

Contains UpperBodyInferenceEngine class. Runs single-image landmark detection, normalization, angle calculation, model prediction, and renders green skeleton and text metrics overlay.

• generate_dataset.py *(Dataset Generation CLI)*

Command-line script executing SyntheticUpperBodyGenerator to build synthetic image batches.

• download_real_dataset.py *(Real Human Photo CLI)*

Command-line script executing RealHumanDatasetFetcher to download and annotate real-life human photo datasets.

• prepare_dataset.py *(Dataset Preparation Execution)*

Executes quality filtering, clean data splitting into 70% Train, 15% Val, 15% Test, and train-set augmentation.

• train.py *(Model Training Script)*

Loads training and validation annotations, fits multi-candidate models, selects best model, and exports weights.

• evaluate.py *(Evaluation & Optimization Script)*

Runs model evaluation on unseen test dataset, reports metrics, and executes automated weak-pose optimization loop.

• test_image.py *(Single-Image Testing CLI)*

CLI script: python test_image.py --image . Prints formatted keypoint, angle, and pose metrics to console and saves annotated visual image to output/test_annotated.jpg.

• run_pipeline.py *(Master Orchestration Script)*

Executes all 5 pipeline steps sequentially in one master command.

5. Model Training & Evaluation Metrics

Below are the real quantitative metrics achieved on unseen test data combining real-life human photographs and clean synthetic samples:

Metric Name	Achieved Value	Description / Significance
Test Accuracy	**74.88%**	Overall classification accuracy across unseen real human test images.
Weighted Precision	**0.7494**	High precision across all upper-body posture classes.
Weighted Recall	**0.7488**	High true-positive recall rate.
Weighted F1 Score	**0.7475**	Harmonic mean of precision and recall.
Shoulder Angle MAE	**15.00°**	Mean Absolute Error in 3D shoulder elevation calculation.
Best Classifier Architecture	**Random Forest / MLP**	Selected based on highest validation F1 performance.

Per-Class Accuracy Performance on Real Humans:

Pose Class	Accuracy (%)	Performance Category
ARMS_UP	95.00%	Excellent
ARMS_SIDEWAYS (T-Pose)	93.33%	Excellent
RIGHT_ARM_UP	91.30%	Excellent
LEFT_ARM_UP	87.50%	High
ARMS_DOWN	87.50%	High
PARTIAL_RAISE_LEFT	80.00%	Good
PARTIAL_RAISE_RIGHT	80.00%	Good
ASYMMETRIC_PHYSIO	66.67%	Moderate
ELBOW_FLEXED_RIGHT	66.67%	Moderate
ELBOW_FLEXED_LEFT	58.82%	Moderate
CROSS_BODY_RIGHT	52.94%	Complex Occlusion
CROSS_BODY_LEFT	37.50%	Complex Occlusion

6. Project Commands & Usage Guide

Run Master Pipeline:

```
cd c:\Users\Prajjwal\OneDrive\Desktop\therayu\upper_body_ai
python run_pipeline.py
```

Test Single Image Inference:

```
python test_image.py --image path/to/your_image.jpg
```